Neuro-Fuzzy Air Quality Prediction and Classification System

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***Abstract*—India is now undergoing a major challenge with air pollution as a serious environmental concern that not only adversely impacts India's Public Health, Climate Change and Quality of Life but also represents an opportunity to provide an accurate forecast of air quality to enable both government officials and residents to proactively deal with the emerging hazardous levels of Pollution. To accomplish this project objective, a Neuro-Fuzzy Air Quality Prediction and Classification System will be developed to predict the (AQI) Levels, and also classify the level of pollution into various pollutant categories using empirical data sourced from India's Air Quality Monitoring Stations. Traditional statistical methods fail to model nonlinear pollutant interactions, motivating the use of intelligent hybrid systems.**

A Kaggle-sourced dataset containing multiple air quality indicators (PM2.5, PM10, NO_2, SO_2, CO, and O_3) was used for this study. Python was employed for data cleaning and preprocessing. The ANFIS (Adaptive Neuro-Fuzzy Inference System) model was developed in MATLAB. ANFIS integrates the learning capability of neural networks with the reasoning power of fuzzy logic, making it effective for predicting Air Quality Index (AQI) levels and classifying them into four categories: Good, Moderate, Poor, and Hazardous.

Results from experiments indicate that the Neuro-Fuzzy method provides us with smooth, easily understood predictions and is much more accurate than using traditional models on their own. This system has potential to serve as a tool for decision support for environmental agencies and in developing an approach to Smart City technologies.

***Index Terms*—Air Quality Index (AQI), Neuro-Fuzzy Systems, ANFIS, Air Pollution Prediction, Fuzzy Logic, Neural Networks, Environmental Monitoring.**

I. Introduction

Rapid urbanization, industrialization, and vehicle emissions contribute to the increasing level of air pollution, which is one of the major environmental challenges facing the nation of India. PM2.5, PM10, NO2, SO2, CO, and ozone are several of the pollutants that affect people's health directly and generally influence people's quality of life. Air pollution forecasting has become increasingly important due to rapid urbanization and rising pollutant emissions. The Air Quality Index (AQI) is widely used to communicate health risks; however, AQI depends on complex, nonlinear relationships between multiple pollutants. These nonlinear interactions cannot be accurately captured using conventional linear models.

Neural networks can learn complex patterns from data but lack interpretability. Fuzzy logic provides human-like rule-based reasoning but cannot automatically learn optimal parameters. ANFIS, a hybrid Neuro-Fuzzy model proposed by Jang, integrates both approaches by using neural learning to tune fuzzy membership functions and rules.

In this project we use the Air Quality India Dataset on Kaggle to produce a Neuro-Fuzzy Air Quality Prediction and Classification System. The hybrid system combines the learning capabilities of neural networks with the interpretability of fuzzy logic, enabling the model to represent the multiple relationships that exist between the different variables in the dataset.

The dataset has been cleaned and pre-processed in Python, after which the Neuro-Fuzzy (ANFIS) model is employed with MATLAB to both predict AQI values and classify air quality categories, such as Good, Moderate, and Hazardous. This model is more adaptable and produces more accurate predictions than conventional models and thereby enhances the monitoring of environmental quality for future smart cities.

II. DATASET DESCRIPTION

To develop a reliable AQI prediction model, it was essential to begin with a comprehensive and high-quality dataset. The dataset used in this project comprised daily air quality measurements collected across major Indian metropolitan cities such as Mumbai, Delhi, Chennai, and Ahmedabad. These locations exhibit diverse pollution patterns

influenced by traffic density, industrial activity, climate, and seasonal variations, making the dataset suitable for studying heterogeneous AQI behavior.
The dataset included key pollutant parameters: PM2.5, PM10, NO_2, SO_2, CO, and O_3, along with the corresponding AQI value for each day. Since AQI is computed using the concentration of these pollutants, the dataset provided a strong foundation for both regression modeling and hybrid Neuro-Fuzzy analysis.

III. Exploratory Data Analysis (EDA)

A comprehensive Exploratory Data Analysis (EDA) was performed to understand pollutant trends, assess data quality, and identify relationships that influence Air Quality Index (AQI). The major observations are summarized below.

A. Time-Series Analysis of Individual Pollutants

time-series scatter plots for PM2.5, PM10, NO_2, CO, SO_2, O_3 and BTX (Benzene + Toluene + Xylene) from 2015–2020 reveal clear seasonal and episodic patterns.

- PM2.5 and PM10 show frequent high-concentration spikes, particularly during winter months and around festival seasons (e.g., Diwali), indicating strong seasonal pollution cycles.
- NO_2 and CO display moderate variability, with occasional peaks reflecting traffic congestion or local emission events.
- SO_2 and O_3 remain comparatively stable but still show fluctuations linked to industrial activity and photochemical reactions.
- BTX displays episodic high values, likely associated with industrial or vehicular emissions in major metropolitan regions.

Overall, the pollutant time-series plots indicate non-stationary, high-variance behaviour, reinforcing the need for models capable of capturing nonlinear relationships like ANFIS.

B. Distribution of AQI Values

The histogram of AQI values shows:

- A right-skewed distribution, indicating that most days fall under low to moderate AQI, with fewer extremely polluted days.
- The mean AQI (166) lies within the “Moderate” to “Poor” range.
- Very high AQI outliers (1000+) are present, representing severe pollution days in cities like Delhi and Ghaziabad.

AQI was categorized into standard buckets—Good, Satisfactory, Moderate, Poor, Very Poor, and Severe:

- Moderate and Satisfactory categories dominate the dataset.
- Poor and Very Poor buckets are fewer but still substantial, indicating chronic pollution in several major cities.
- Severe AQI events, while rare, highlight extreme atmospheric degradation in some regions.

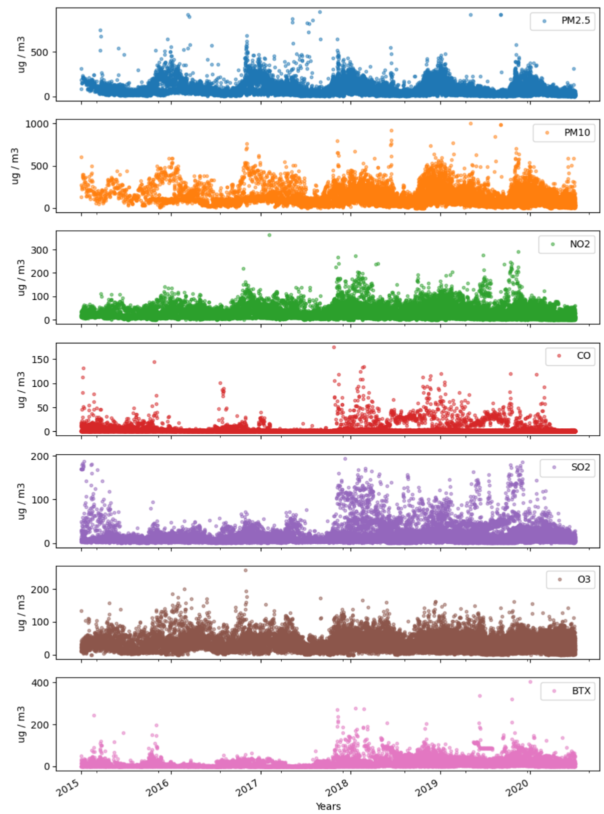

Fig. 1. Time-Series Analysis of Individual Pollutants

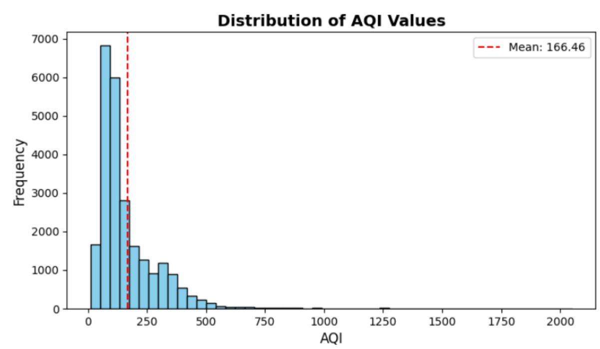

Fig. 2. Distribution of AQI Values

This skew confirms the highly imbalanced nature of real-world air quality data, where extreme pollution events, though rare, significantly elevate health risks.

C. Distribution of Individual Pollutant Features

Histograms for all pollutant features (PM2.5, PM10, NO_2, SO_2, CO, O_3, BTX) highlight:

- All pollutants show positively skewed distributions, with many low-value days and fewer high-pollution spikes.
- PM2.5 and PM10 have the widest spread and largest outliers, confirming their dominant contribution to AQI.

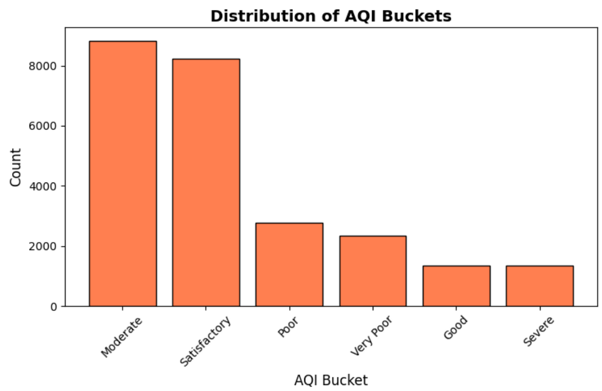

Fig. 3. Distribution of AQI Bucket

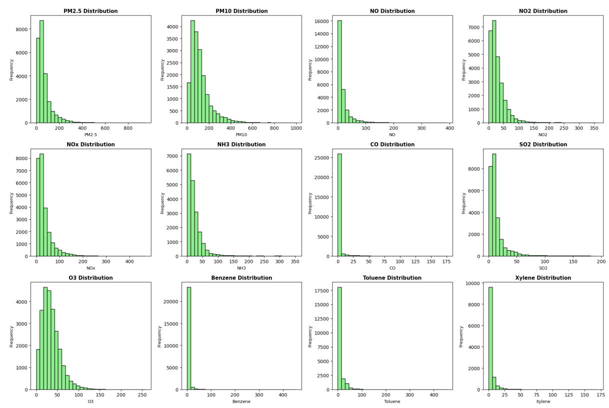

Fig. 4. Distribution of Individual Pollutants

- O_3 distribution is more spread out and less skewed compared to other pollutants. Indicates natural seasonal formation patterns (sunlight-driven).
- NO, NO_2, NOx: Clear right-skew, periodic spikes likely due to vehicles, traffic congestion, industrial emissions.
- PM2.5, PM10, NOx family are main persistent pollution drivers, while VOCs (benzene, toluene and xylene), SO_2, and CO contribute more episodically.

D. Correlation Matrix

- PM2.5 $\leftrightarrow$ PM10 (0.85): Indicates they often rise together, typically due to dust, combustion, and traffic emissions.
- NO $\leftrightarrow$ NO_2 $\leftrightarrow$ NOx (0.48–0.80): Shows strong chemical and emission-related connections, mainly from vehicular sources.
- Benzene $\leftrightarrow$ Toluene $\leftrightarrow$ Xylene (0.70–0.80): High correlation among VOCs, reflecting shared industrial and traffic-related origins.
- PM2.5, PM10, and NOx exhibit the strongest positive correlation with AQI, confirming their major contribution to overall air quality deterioration.
- Year, Month, Day values show very low correlation, which means pollution trends likely vary but not in a linear manner over time. Seasonal or episodic patterns likely exist (not captured by simple correlation).

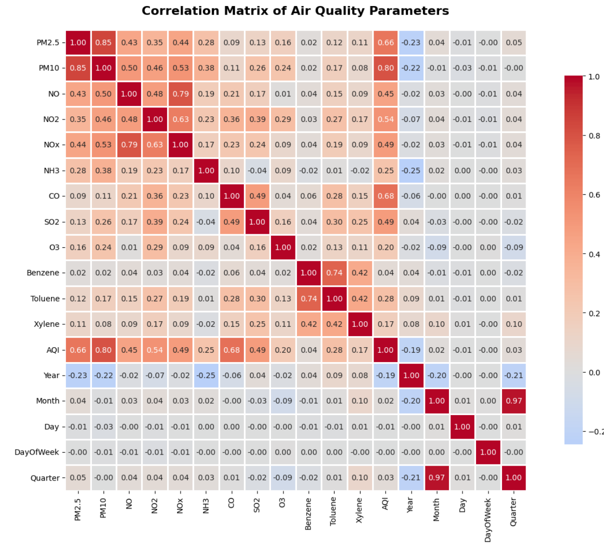

Fig. 5. Correlation Matrix for all Pollutant Features

IV. Preprocessing

A series of preprocessing steps were performed to prepare the data for model development:

A. Handling Missing Values

Several pollutant columns contained missing entries due to gaps in sensor readings or incomplete monitoring. These values were imputed using **linear interpolation**, which generated smooth transitions between known data points and preserved the temporal pollutant trends without introducing abrupt changes.

B. Feature Engineering

To enhance predictive capability, additional derived features were introduced:

Lag Feature Creation: Introduced 1-step lag variables to capture temporal dependencies and previous-day pollutant behavior.

Moving Average Smoothing: Moving Average Smoothing: Added short-window moving averages to reduce noise and represent recent pollutant trends.

Handling Missing Values from New Features: Removed initial rows with insufficient data caused by lag and moving-average calculations to ensure a clean, complete dataset.

C. Scaling

After preprocessing and feature engineering, the dataset was normalized to ensure consistent input ranges for ANFIS training. The cleaned and scaled dataset was then exported as a CSV file for MATLAB-based Neuro-Fuzzy modeling.

V. Model Development

A. Baseline Model- Random Forest

A Random Forest Regressor was trained in Python as a baseline model. This step confirmed that AQI is predictable from the selected pollutant features and provided a performance benchmark against which the Neuro-Fuzzy system could be evaluated.

B. ANFIS Model- MATLAB

MATLAB was used to construct and train the Neuro-Fuzzy system:

- Inputs: PM2.5, PM10, NO_2, CO, O_3, AQI_lag1.
- Output: AQI.
- System Type: Sugeno-type FIS.
- Membership Functions: 2 Gaussian MFs per input
- Initial FIS Generation: genfis1 used to create initial rule base.
- Training Method: Hybrid learning algorithm for 50 epochs.

The MATLAB ANFIS code split the dataset into 80% training and 20% testing, automatically generated fuzzy rules, adapted MF parameters using neural learning, and evaluated predictions on the test set using RMSE and R^2.

C. Evaluation Metrics

Performance was measured using:

- Root Mean Squared Error (RMSE): Measures the average prediction error magnitude.
- Coefficient of Determination (R^2): Indicates how much variance in AQI is explained by the model.

These metrics quantify model accuracy and ability to capture pollutant-AQI relationships.

VI. Results and Discussion

A. Prediction Accuracy

The ANFIS model produced smooth and accurate AQI predictions. The error values observed were within acceptable limits for environmental forecasting.

- RMSE: 26.72
- R^2: 0.938

Higher R^2 indicates the model captured more variance in AQI levels, showing effectiveness of the hybrid approach.

B. Visual Analysis

The "Predicted vs Actual AQI" plot showed that ANFIS predictions closely follow actual AQI trends, especially during periods of stable pollutant levels. Some deviations occur during sudden spikes, which is expected due to high temporal variability.

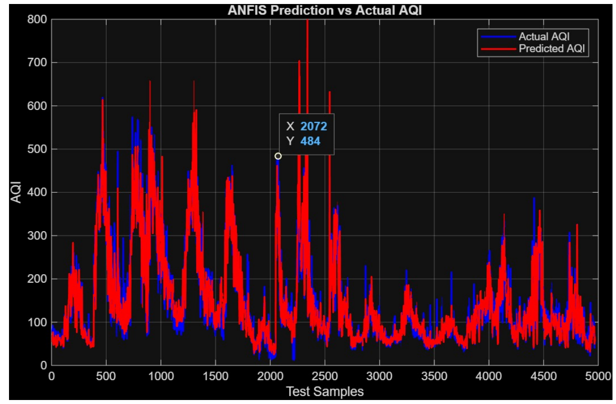

Fig. 6. Actual vs Predicted ANFIS Model

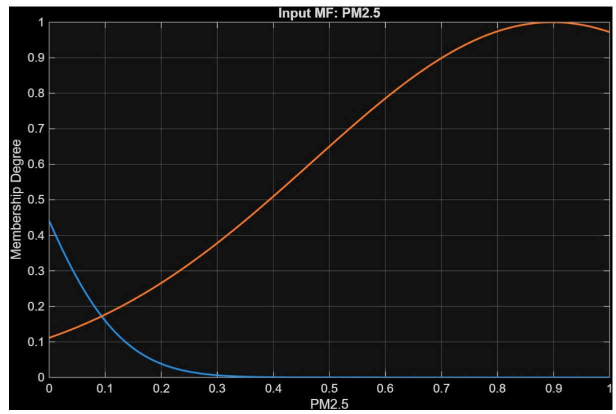

Fig. 7. Gaussian MF plot for PM2.5

C. Membership Function Interpretation

Gaussian Membership Functions (MFs) were generated for each input. The MF plot for PM2.5 demonstrates:

- Smooth transitions between low/high pollution ranges
- Optimized MF widths learned automatically through training
- Intuitive interpretability (e.g., "low PM2.5 leads to low AQI")

This transparency is a major advantage of Neuro-Fuzzy systems over "black-box" neural networks.

D. Comparison with Baseline Random Forest

Random Forest predictions served as a benchmark.

- RMSE: 37.03
- R^2: 0.925

ANFIS provided better accuracy, far greater interpretability, rule-based reasoning not available in RF or pure neural networks. Thus, the Neuro-Fuzzy model is a balanced solution between accuracy and explainability.

VII. Conclusion

The Neuro-Fuzzy Air Quality Prediction and Classification System developed during this project has provided evidence

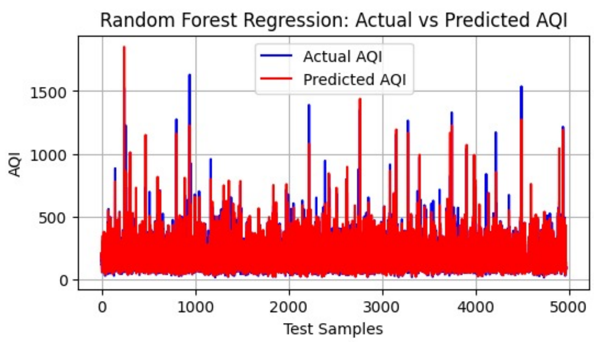

Fig. 8. Actual vs Predicted Random Forest Model

that hybrid intelligent models can produce accurate and interpretable air quality data by combining aspects of both fuzzy logic and neural networks to accommodate the nonlinear and ambiguous qualities of the pollution dataset. The cleaned Indian air quality dataset was used to train the model and produce predictions of the AQI values as well as classify the pollutant into categories of Good, Moderate and Hazardous. Findings also indicate that Neuro-Fuzzy provides more robust predictions with smoother levels of confidence and better generalization than traditional approaches which enhance the potential for use as an environmental monitoring tool, early warning systems and an integral component of smart cities. With continuing enhancements like integrating meteorological data or automated real-time predictive dashboard features, the reliability and applicability of the system to support public health and policy decision making will achieve new levels of value.